WEEK 12

COLOR MEMORY NOTES

IoT Data Analytics + Machine Learning + Big Data + Security

Fast revision for tomorrow's exam

COLOR CODE

BLUE	Core definition / concept
GREEN	Feature / use / benefit
YELLOW	Must-memorize fact
ORANGE	Comparison / exam clue
PURPLE	Memory trick
RED	Very high priority

Emergency Priority

1) DDPP analytics 2) Scaling + Volatility 3) Local vs Remote

4) MBOS ML domains 5) Big Data 3V 6) IoT Security 8 challenges

7) Structured vs Unstructured 8) CIPAV flow

1. MASTER EXAM MAP

Use the clue to identify the answer

What is happening?	->	**Descriptive**
Why did it happen?	->	**Diagnostic**
What is likely to happen?	->	**Predictive**
What should I do?	->	**Prescriptive**
Data processed at sensor/gateway	->	**Local Learning**
Data sent to data center/cloud	->	**Remote Learning**
Huge data size	->	**Volume**
Many data formats	->	**Variety**
Fast data arrival	->	**Velocity**
Threshold reached -> action	->	**Behavior Control**

MASTER CHAIN

MBE -> DDPP -> CIPAV -> SV -> Local/Remote -> MBOS -> 3V -> Security 8

2. IoT Data Analytics - Core Areas

The slide groups analytics around data, ML, Big Data and Edge

CORE IDEA

For IoT analytics, data is the central subject. The slide highlights Machine Learning, Big Data Analytics Tools and Technology, and Edge Streaming Analytics.

IMPORTANT POINTS

- Machine Learning: gain business insights from collected data.
- Big Data tools mentioned: Hadoop, NoSQL, MapReduce and MPP.
- Edge Streaming Analytics: process and analyze data close to the endpoint.
- Real-time processing is important in IoT.

MEMORY

MBE = Machine Learning - Big Data - Edge Streaming

BIG DATA TOOL MEMORY

HNMM = Hadoop - NoSQL - MapReduce - MPP

3. Data Types

Structured vs Unstructured + Data in Motion vs Data at Rest

STRUCTURED

- Organized format
- Conforms to a model
- Easy tabular representation
- Examples: spreadsheets, databases

UNSTRUCTURED

- No fixed model
- Not organized like a table
- Examples: text
- Images, video and speech

DATA IN MOTION vs DATA AT REST

The W12 slide introduces the contrast as an IoT data concept. For exam memory, think 'in motion' = actively moving/streaming data and 'at rest' = stored data. The slide itself provides the heading but no detailed bullet explanation.

MEMORY

Structured = Table | Unstructured = Media

4. Four Types of IoT Data Analytics

Most important diagram in W12

Descriptive	What is happening?
Diagnostic	Why did it happen?
Predictive	What is likely to happen?
Prescriptive	What should I do about it?

ANALYTICS MEMORY

DDPP = Descriptive - Diagnostic - Predictive - Prescriptive

QUESTION MEMORY

What -> Why -> Future -> Action

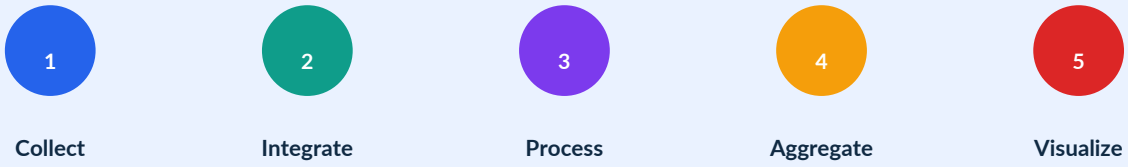
5. Value, Complexity and Analytics Flow

As analytics becomes advanced, value and complexity rise

VALUE + COMPLEXITY ORDER

1. Descriptive	starting level
2. Diagnostic	more advanced -> more value + complexity
3. Predictive	more advanced -> more value + complexity
4. Prescriptive	more advanced -> more value + complexity

BIG DATA TECHNOLOGY FLOW



FLOW MEMORY

CIPAV = Collect - Integrate - Process - Aggregate - Visualize

6. IoT Data Analytics Challenges

Two explicit challenges in the slide

SCALING PROBLEM

- Many smart objects continually send data.
- Relational databases can grow very quickly.
- Performance issues may appear.
- Fixes may require more hardware.
- Architecture changes can be costly.

DATA VOLATILITY

- Traditional relational schema should be designed carefully.
- Changing schema later can slow/stop operations.
- IoT data models evolve over time.
- Dynamic schema may be required.
- Changes may be needed daily or hourly.

CHALLENGE MEMORY

SV = Scaling - Volatility

FAST CLUE

More devices -> database growth = Scaling | Data model keeps changing = Volatility

7. Machine Learning and IoT

Supervised/Unsupervised + Local/Remote learning

LEARNING TYPES MENTIONED

- Supervised Learning
- Unsupervised Learning
- Local Learning
- Remote Learning

LOCAL LEARNING

- Data collected locally
- Processed locally
- Sensor itself = edge node
- Gateway = fog node

REMOTE LEARNING

- Data collected at devices
- Sent to central unit
- Processed in data center
- Or processed in cloud

MEMORY

Local = Here (Edge/Fog) | Remote = There (Data Center/Cloud)

8. Four Major ML Application Domains

Monitoring -> Control -> Optimize -> Self-Heal

Monitoring	Observe environment/operation and detect abnormal or early-failure conditions.
Behavior Control	When thresholds are reached, generate alarms or trigger corrective actions.
Operations Optimization	Use analytics to improve overall efficiency, not only correct faults.
Self-Healing / Self-Optimizing	System learns, finds new optimizations, detects defects and can self-heal.

ML DOMAIN MEMORY

MBOS = Monitoring - Behavior Control - Operations Optimization - Self-Healing

9. ML Domain Examples

Turn slide examples into exam-ready memory

MONITORING + BEHAVIOR CONTROL

- Monitoring examples: temperature, humidity, CO2, pump pressure, oil viscosity.
- ML can detect out-of-range behavior and early failure conditions.
- Behavior control may generate an alarm when a threshold is reached.
- Advanced systems can trigger corrective action automatically.

OPTIMIZATION + SELF-HEALING

- Optimization aims to improve the overall process and efficiency.
- Example: choose better chemical/temperature/stirring combinations in a purification plant.
- Self-optimizing systems can dynamically test and combine parameters.
- They can implement new optimizations when gains are detected.
- They can detect new deviations and pre-detect possible defects.

FAST MEMORY

Observe -> Act -> Improve -> Heal

10. Big Data 3V

Very easy marks - Speed, Type, Size

VELOCITY

Speed

How quickly data is collected and analyzed.

VARIETY

Type

Different data types: structured, semi-structured, unstructured.

VOLUME

Size

Scale/amount of data, from GB to PB or even Exabyte.

3V MEMORY

Velocity = Speed | Variety = Type | Volume = Size

11. IoT Security Challenges

Final slide - 8 challenges

8 SECURITY CHALLENGES

- Inconsistent security standards
- Low processing power
- Legacy assets
- Lack of awareness in users
- Botnet attacks
- Lack of encryption
- Missing firmware updates
- Rogue and counterfeit IoT devices

EASIER GROUPING

DEVICE: low power, legacy, missing updates, rogue devices | HUMAN/STANDARD: inconsistent standards, low awareness | ATTACK/PROTECTION: botnet, lack of encryption

12. Final 5-Minute Revision

Read this page 3 times before the exam

FACT BANK

- MBE = Machine Learning - Big Data - Edge Streaming
- Structured = organized model; Unstructured = text/image/video/speech
- DDPP = Descriptive - Diagnostic - Predictive - Prescriptive
- Questions = What - Why - Future - Action
- Value + complexity rise toward Prescriptive
- CIPAV = Collect - Integrate - Process - Aggregate - Visualize
- SV = Scaling - Volatility
- Local = Edge/Fog; Remote = Data Center/Cloud
- MBOS = Monitoring - Behavior Control - Operations Optimization - Self-Healing
- 3V = Velocity - Variety - Volume
- Velocity = Speed | Variety = Type | Volume = Size
- Security 8: standards, low power, legacy, awareness, botnet, encryption, firmware, rogue devices

ULTIMATE MEMORY CHAIN

MBE -> DDPP -> CIPAV -> SV -> Local/Remote -> MBOS -> 3V -> Security 8

13. Top Questions + 8-Mark Formula

What to prepare first tonight

TOP 9 QUESTIONS

- Explain four types of IoT Data Analytics.
- Compare Structured and Unstructured Data.
- Explain IoT Data Analytics challenges.
- Explain Local vs Remote Learning.
- Explain four ML application domains in IoT.
- Explain Big Data 3V.
- Explain IoT Security challenges.
- Explain the IoT Data Analytics overview / CIPAV flow.
- Explain value and complexity of analytics types.

8-MARK ANSWER FORMULA

- 1. Start with one-line definition.
- 2. Name the exact type/topic.
- 3. Write 4-8 clean syllabus points.
- 4. Use slide keywords such as threshold, schema, edge/fog, cloud, 3V.
- 5. Add one small comparison/example if available.
- 6. Finish with a short conclusion.

LAST RULE

Memorize keyword order, not full paragraphs.